ASSEMBLYAI VOICE AGENT HACKATHON

Out Loud

Courseware you operate by voice.

Team Out Loud · Solo entry · Built on the AssemblyAI Voice Agent API

out-loud-voice-tutor-x18c.vercel.app

THE PROBLEM

Everyone clicks through mandatory training.

Next. Next. Next. Submit. The completion certificate proves attendance, not learning — and everybody involved knows it.

Voice tutors that already exist turn your material into something that talks about it. A podcast of a course is still not the course.

Nothing so far makes the learner do anything.

What actually changes

The learner has to produce the answer,

out loud, before anything moves.

Speaking is the input device —

not a narration layer on top of one.

THE IDEA

You take the course by talking to it.

Ada reads you a real workplace situation — a graduate chosen to lead longer-serving colleagues, a role described as better suited to someone younger.

You answer out loud. She puts your answer into the screen for you: files the card, moves the character, selects the option.

Fourteen screens, three parts, one continuous conversation.

THE RULE IT IS BUILT ON

Ada can answer for you.
She can never move you on.

That is not a line in a prompt. It is enforced by the tool list: there is no tool that presses Submit, Continue or Next.

A prompt rule is a request. A missing tool is a guarantee.

It runs the other way too — a screen is read in full the first time before anyone goes anywhere. The course can never run ahead of the person taking it.

WHAT MAKES IT DIFFERENT

She operates live courseware.

By pressing the same controls a hand would press. The course does not know it is being driven by voice.

01

Sorting board

“That one's a reasonable adjustment.” The card files itself into the column.

02

Platformer game

Name the block and the pixel character runs across the level and lands on it.

03

Quiz

Say the answer. The real option is selected, scored and carried into your total.

HOW IT IS BUILT

One socket. Eight tools.

A single AssemblyAI Voice Agent API WebSocket does speech-to-text, reasoning and speech back — no stitched-together pipeline.

Eight JSON-schema tools give the agent eyes and hands on the page: read the screen, answer the activity, navigate where asked.

An Express backend mints short-lived, single-use tokens, so the API key never reaches the browser.

AudioWorklet streams 24 kHz PCM16 in both directions, with scheduled playback so speech stays continuous. Deployed on Vercel.

Stack

AssemblyAI Voice Agent API

Node.js + Express

Vanilla JS, no framework

AudioWorklet / Web Audio

Vercel

THE LEARNING DESIGN

Every part closes with a named technique.

Evidence-backed, and told to the learner by name so they can use it again tomorrow.

PART 1

Elaborative interrogation

Why would that be true here?

PART 2

Keyword mnemonic

One handle for the whole rule.

PART 3

Spaced retrieval

Pulled back out, later, cold.

WHY THIS SHOULD WORK

Intelligent tutoring is already close to human tutoring.

VanLehn (2011) puts step-based intelligent tutoring at d = 0.76 against d = 0.79 for human tutors — near-parity, at software cost.

Out Loud takes the part that stayed stubbornly human — being made to say it, to someone, before you move — and gives it to the machine.

WHAT WAS BUILT WHEN

Declared up front.

BEFORE

Prior work

The underlying 14-screen module — its scenes, activities and visual design — is my own, built before the hackathon.

DURING

Built in the window

The agent, the eight tools, the token server, the audio pipeline and every spoken activity mechanic.

TRY IT YOURSELF

Out Loud

Live app

out-loud-voice-tutor-x18c.vercel.app

Code

github.com/KV2699/out-loud-voice-tutor

Built solo on the AssemblyAI Voice Agent API